

Reporte de Análisis Físico

Validación MC y Asimetría UMD/SD

Configuración de la Simulación

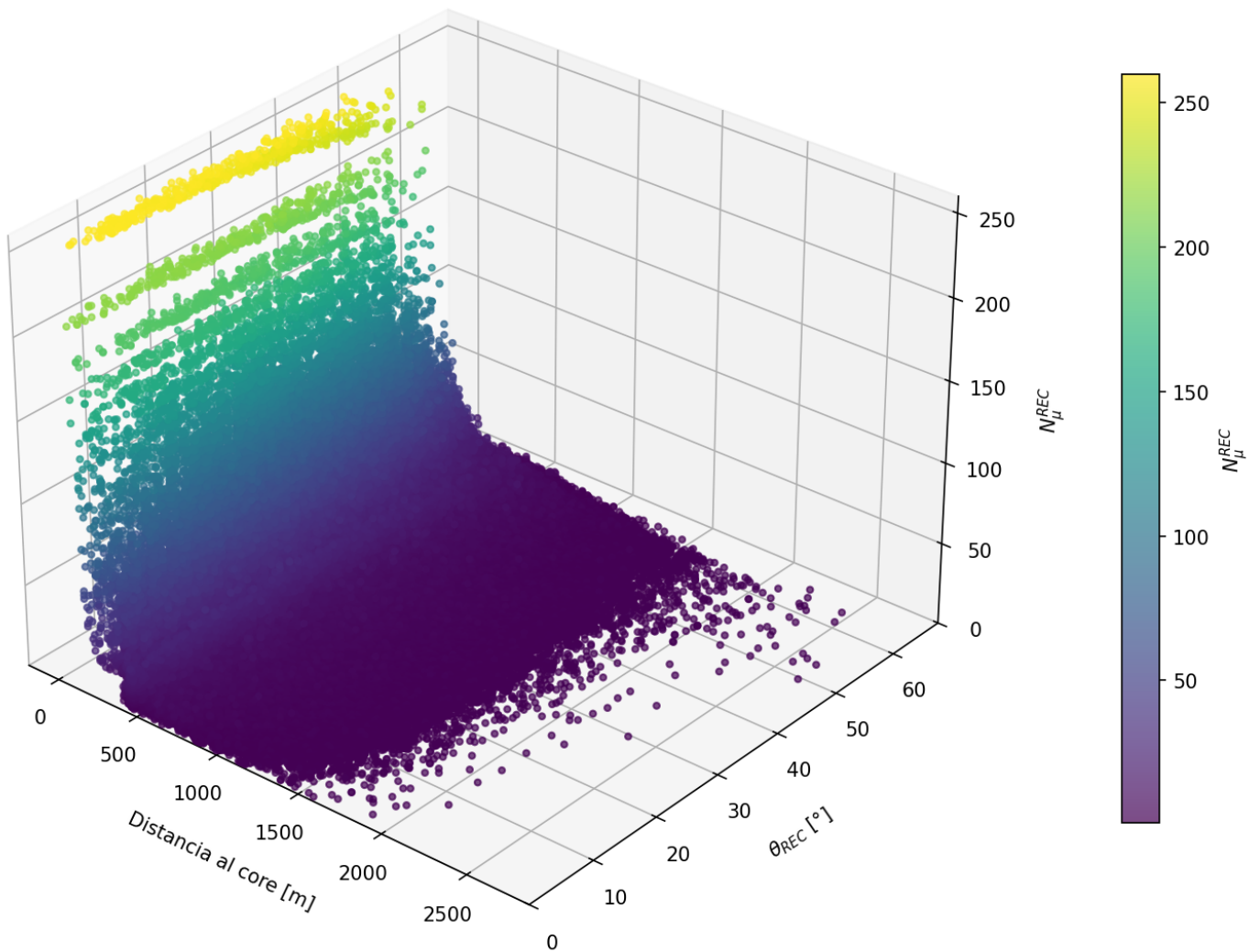
Parámetro	Valor
Modelo Hadrónico	SIBYLL 2.3d
Primario	Oxígeno
Rango Energía	17.5 - 18.0 log(eV)
Carpeta	
Eventos Totales	4,999
Módulos Totales	1,649,367

1. Limpieza de Datos

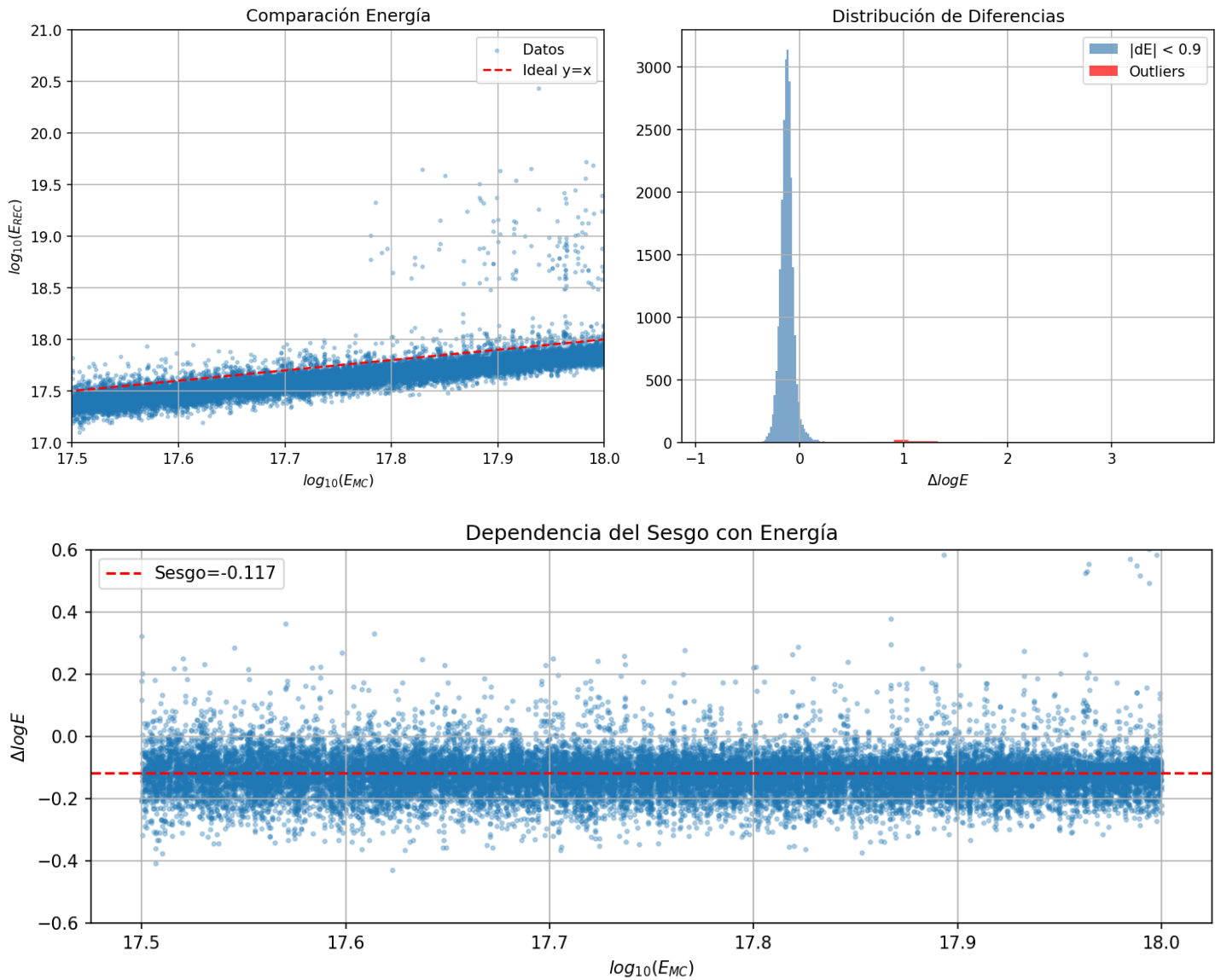
REPORTE DE LIMPIEZA

Métrica	Valor
Filas Iniciales	1649367
Filas Eliminadas	63492 (3.85%)
Filas Finales	1585875
Causa Principal	NaN en logE_REC (63492)

2. Visualización General

N_{μ}^{REC} vs distancia y θ 

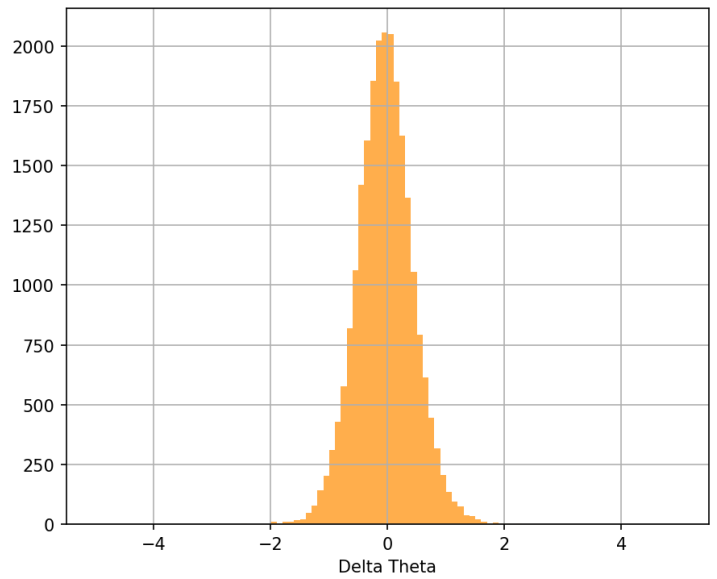
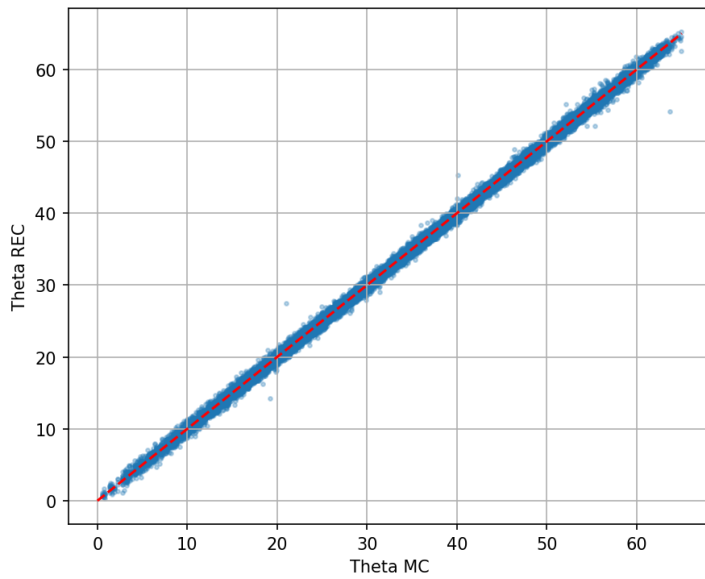
3. Resolución de Energía



Sesgo (Mean): -0.1170
 Resolución (Std): 0.1109
 Outliers (> 0.9): 70 (0.30%)

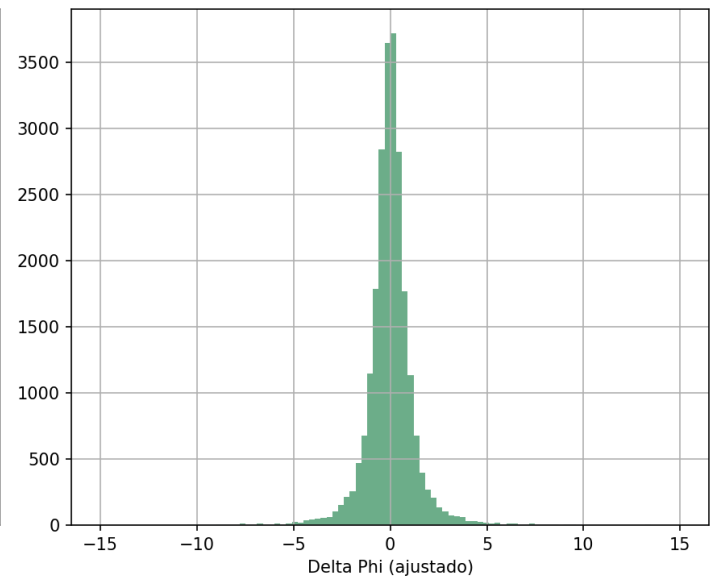
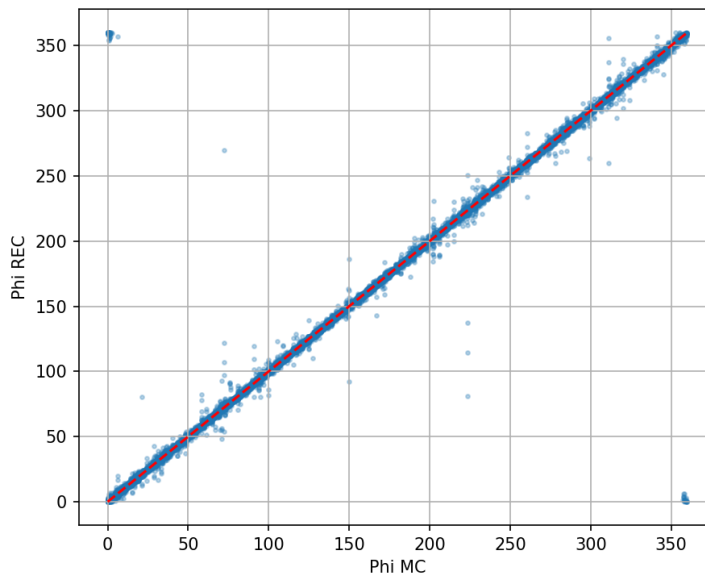
4. Resolución Angular

Theta (Cenital)



Sesgo: -0.0445 deg
 Resolución: 0.4889 deg
 Outliers ($> 10.0^\circ$): 0 (0.00%)

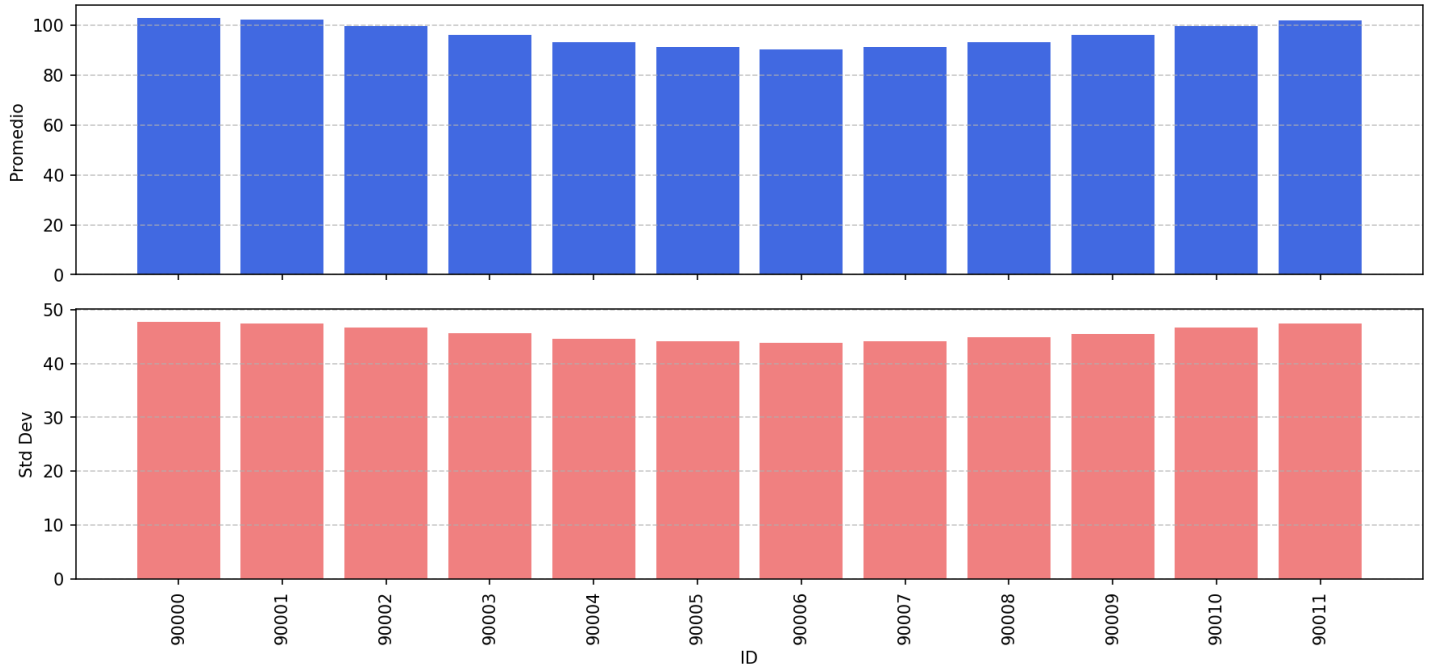
Phi (Azimutal)



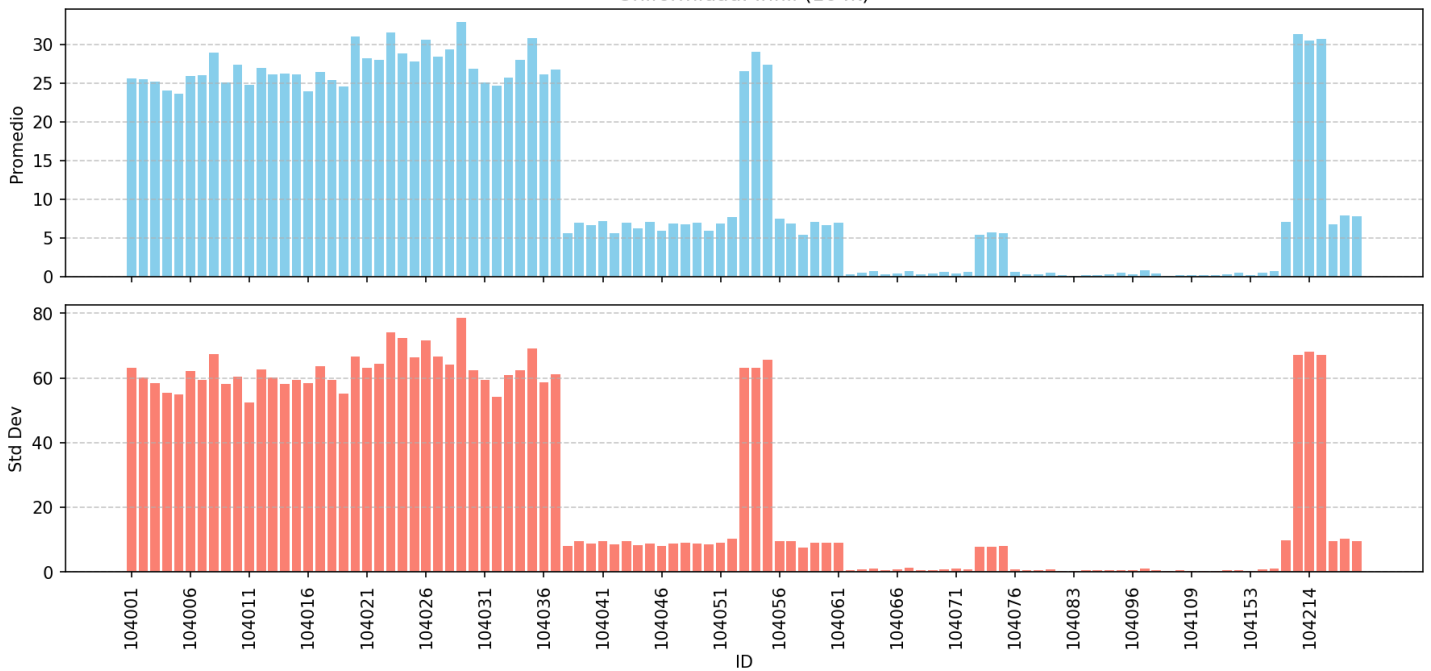
Sesgo (Median): -0.0013 deg
 Resolución (MAD): 0.5180 deg
 Outliers ($> 15.0^\circ$): 50 (0.21%)

5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

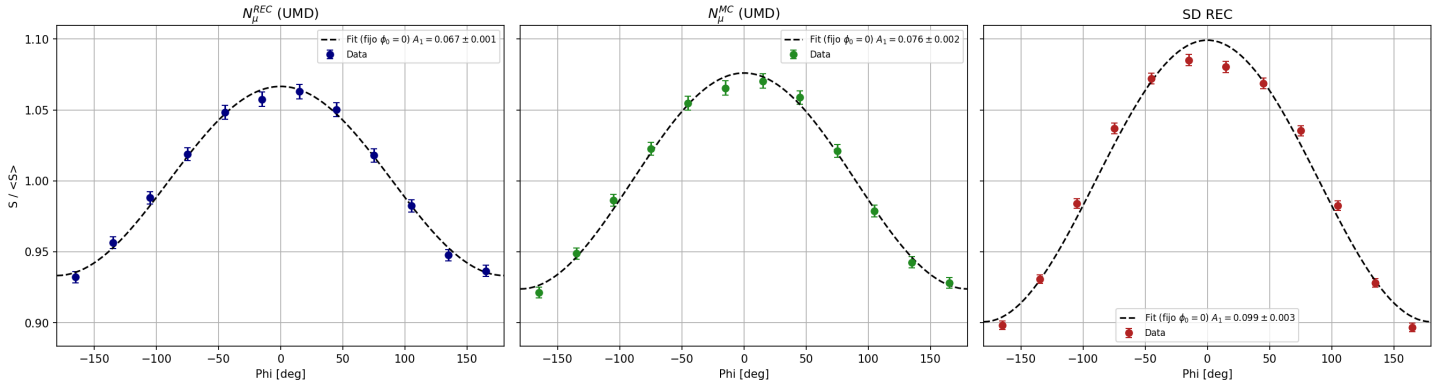
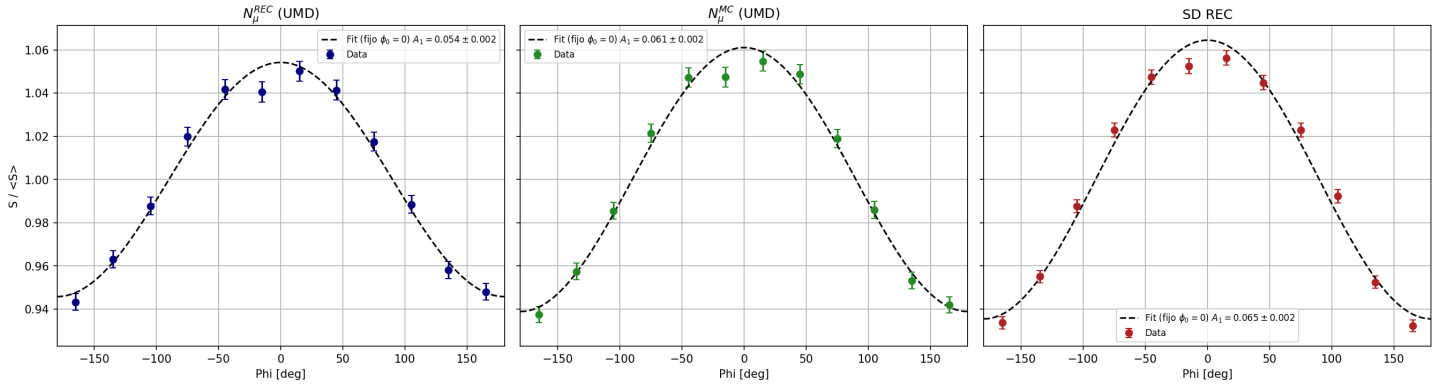
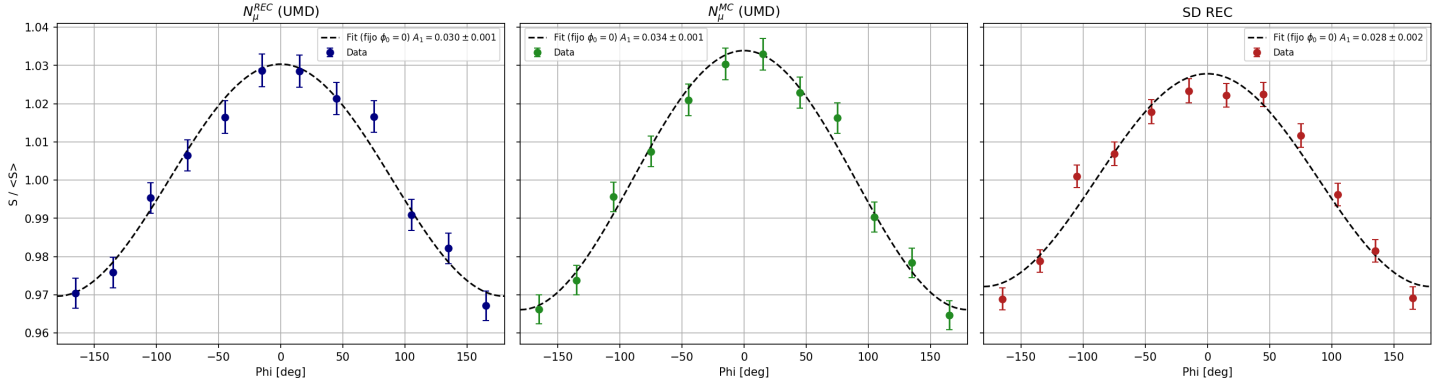


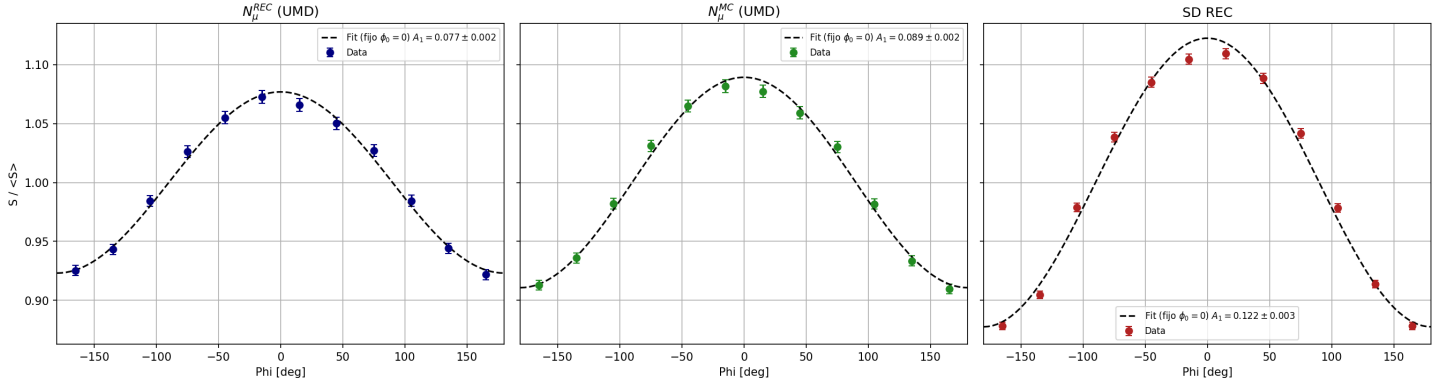
Uniformidad: Infill (104k)



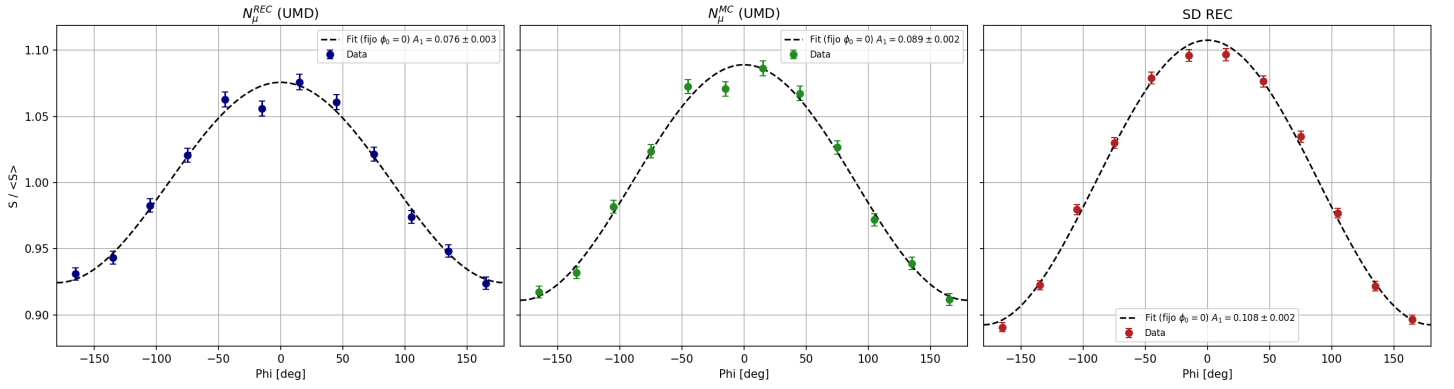
6. Asimetría Azimutal

Bin 0: Theta 0.4 - 21.7 deg

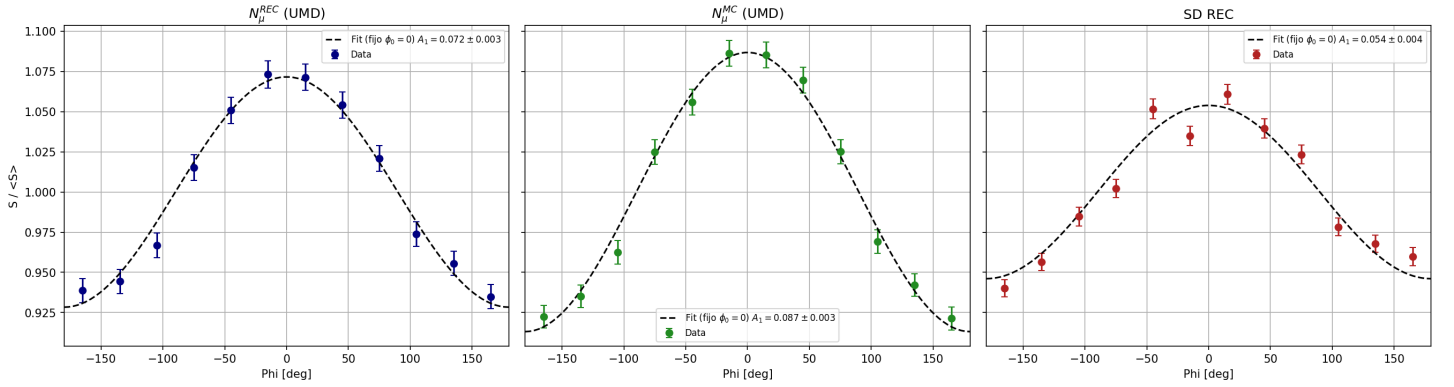




Bin 4: Theta 47.7 - 55.8 deg



Bin 5: Theta 55.8 - 65.0 deg



7. Resumen de Ajustes (Comparación)

Resultados Detallados: Amplitud A1 y Error

theta_range	A1_UMD_REC	Err_UMD_REC	A1_UMD_MC	Err_UMD_MC	A1_SD_REC	Err_SD_REC
0.4-21.7	0.0304	0.0015	0.0339	0.0013	0.0278	0.0017
21.7-31.6	0.0542	0.0019	0.0611	0.0019	0.0645	0.0023
31.6-39.9	0.0667	0.0015	0.0761	0.0017	0.0992	0.0033
39.9-47.7	0.0769	0.0018	0.0893	0.0021	0.1225	0.0031
47.7-55.8	0.0757	0.0026	0.0890	0.0024	0.1076	0.0020
55.8-65.0	0.0716	0.0027	0.0869	0.0028	0.0539	0.0040

